

Supplementary Material for “Unifying Graph Traversal and Time Series Joins in Hybrid Graphs”

This document contains additional plots and their description, omitted from the revision paper due to space constraints for the ICDE 2026 paper entitled “Unifying Graph Traversal and Time Series Joins in Hybrid Graphs”.

1 Evaluation

RQ3 How does query selectivity and the number of matching paths affect the intermediate results size and runtime?

RQ4 How does the subsequence length influence the comparison cost when using z-normalised Euclidean distance?

RQ5 How does a given Allen’s relation impact the number of TD-Join candidate pairs and performance?

1.1 Query selectivity (RQ3)

Both datasets show clear and consistent trends: as selectivity increases, execution times grow steadily across all temporal relations and window sizes. This confirms that higher selectivity directly translates into more temporal paths that need to be computed, which in turn increases query cost. In these datasets, the growth pattern is monotonic, meaning that each increment in selectivity corresponds to higher computational overhead.

However, this monotonic trend does not hold for the *equal* relation in the LA dataset, where execution times do not consistently increase with selectivity. This deviation suggests that other factors, such as time series distribution or number of time series, may influence computational cost for this specific relation.

1.2 Subsequence length (RQ4)

For the METR-LA dataset, an increasing subsequence length w reflects higher computational cost, since the cost of computing the z-normalized Euclidean distance is directly proportional to the length of the considered subsequences.

In the Synthea dataset, this trend is not consistent: although, in general, increasing subsequence length w leads to higher computational cost, this is not always the case. As subsequence length increases, the total number of available subsequences to compare decreases, which can result in lower query times.

1.3 Variability wrt the Allen’s Relation (RQ5).

For both the Synthea and METR-LA datasets, the computational cost of temporal relations depends on the number of subsequence comparisons. In general, more comparisons lead to higher execution times.

The *before* relation is the most expensive, as each subsequence must be compared to all subsequent ones. In Synthea, longer subsequences w can sometimes reduce cost, since fewer candidates satisfy the *before* condition. *Overlaps* is also costly, but less than *before*, with execution time increasing for larger w and selectivity. *Meets* and *equal* are lightweight, requiring only single comparisons; in Synthea, *equal* may show non-monotonic behavior due to the distribution of exact matches.

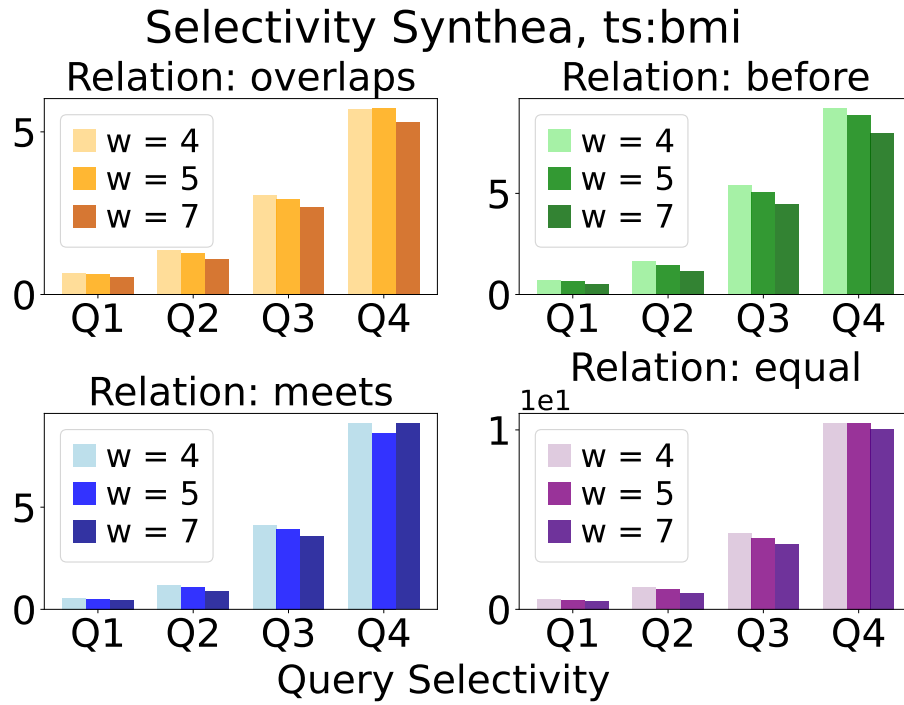


Figure 1: Selectivity analysis with Synthea dataset.

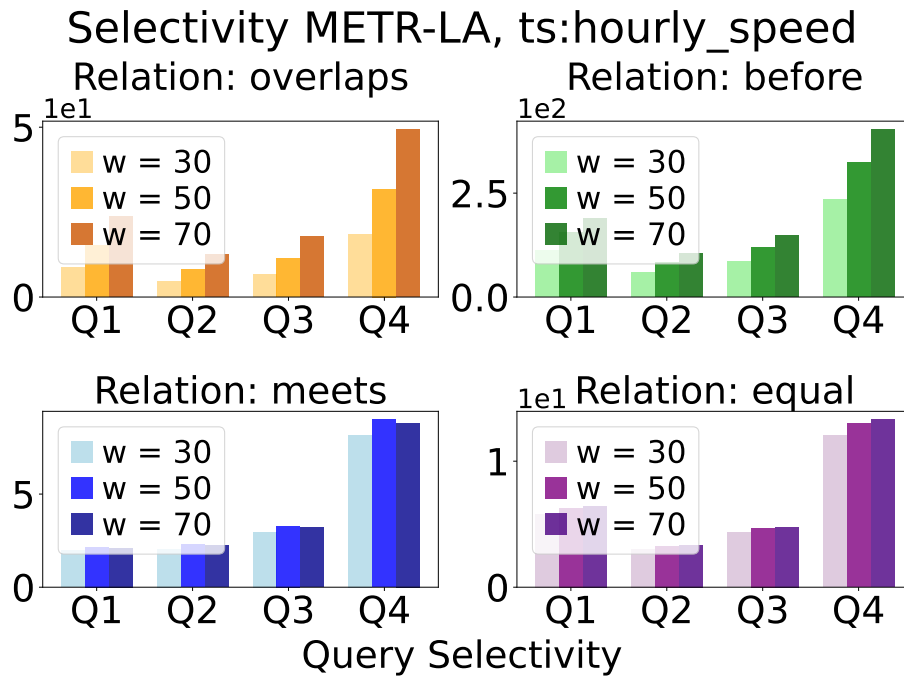


Figure 2: Selectivity analysis with METR-LA dataset.